

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2232

Birchwood 138 kV
October 03, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2232. The Interconnection Customer proposes a 78.0 MW battery energy storage facility interconnecting at the Birchwood 138 kV substation in the Virginia area. The request was received on November 18, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 496 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,413,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ingleside Substation to Blake 138kV line	Nauvoo to Birchwood 345kV	148.7	242	11.1
Ingleside to Comorn Dp 138kV	Birchwood to Nauvoo 345kV line	148.4	245	11.1
South Kirkland to Birchwood 138kV	Nauvoo to Birchwood 345kV L	148.3	279	11.1
S Kirkland to Comorn Dp 138kV	Nauvoo to Birchwood 345kV	133.1	292	11.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Blake to Inglesde 138kV Line	Rebuild line with higher-capacity conductor	\$8,541,000	100% (\$8,541,000)
Ingleside to Comorn Dp 138kV	Rebuild line with higher-capacity conductor	\$14,003,000	100% (\$14,003,000)
South Kirkland to Birchwood 138kV	Rebuild line with higher-capacity conductor	\$5,070,000	100% (\$5,070,000)
Comorn Dp to South Kirkland 138kV	Rebuild line with higher-capacity conductor	\$1,799,000	100% (\$1,799,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$29,413,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.